WHEN LEARNED WARM STARTS HELP: A CONTROLLED STUDY OF SOURCE LINEARITY, PRECONDITIONING, AND COMPLETE SOLVE COST

A REPRODUCIBLE COMPUTATIONAL STUDY

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ABSTRACT

Learned initial guesses can improve prediction error without reducing accepted numerical correction work. We study this distinction in synthetic one-dimensional elliptic systems, preserving a failed quadratic-ensemble predecessor and evaluating a separately frozen source-linear successor. The original strict-tolerance confirmatory iteration reduction is 0.00% on IID cases. In the successor's predeclared 129-point, 10^{-6}-tolerance IID cell, five independent training/evaluation blocks yield 27.73% mean paired iteration reduction against zero-initialized Laplacian PCG and 21.17% against a physics-only initial field. The corresponding descriptive block-bootstrap intervals are [27.28, 28.22]% and [20.94, 21.50]%. Complete online time improves by 21.72% and 26.07%, respectively. The frozen primary rule is satisfied. However, direct tridiagonal solving remains 5.61 times faster by the ratio of mean times. The full successor matrix contains 17280 condition-level rows, with 0 acceptance failures. These findings support only the stated iterative-solver comparison; they do not establish a new solver family, multidimensional superiority, or independent investigator reproduction.

Keywords elliptic equations · learned warm starts · conjugate gradients · reproducibility

1 Introduction

Repeated elliptic solves arise in parameter studies, inverse problems, and design. A learned solution map can supply an initial field to a conventional numerical solver, whose residual criterion then decides acceptance. This division is attractive: prediction need not achieve the final accuracy, while the numerical correction need not begin from zero. Its success, however, depends on the downstream algorithm, problem spectrum, tolerance, and complete computational cost. A small prediction error is not itself evidence of an efficient solver.

Our starting study, S1, fits a quadratic ridge ensemble to a synthetic one-dimensional diffusion family. At relative residual tolerance 10^{-10}, both learned and zero initializations use 63 iterations on every one of the 128 development evaluation cases. The mean paired iteration reduction is exactly zero in both the in-distribution and coefficient-shifted splits. This result motivates diagnosis rather than deletion or a retrospective endpoint change. S1's original configuration and raw artifacts are preserved, and its confirmatory configuration is evaluated without retuning.

We formulate a separately versioned successor, S2, around an exact structural property: for a fixed diffusion coefficient and homogeneous boundaries, the solution depends linearly on the forcing. An unconstrained quadratic map of coefficient and forcing coordinates contains terms inconsistent with that property. We therefore fit a source-linear coefficient-polynomial map, test it against an unconstrained regression control, and separate the effect of the starting field from a strong classical preconditioner.

The contribution is a compact, auditable computational study. It supplies explicit source identities, preserved failures, complete case-level evidence, and an executable comparison of accuracy, iteration count, and complete online time. It does not introduce a new family of hybrid solvers, prove a universal learned speedup, or establish superiority

on realistic multidimensional PDEs. In one dimension a direct tridiagonal algorithm is an especially important alternative; a learned iterative method must justify itself against that inexpensive baseline.

2 Related work and novelty boundary

HINTS combines neural operators and classical relaxation using complementary spectral behavior [1]. NOWS directly studies neural-operator initial guesses for Krylov methods [2]. These precedents rule out treating learned initialization alone as our novelty. We do not reproduce their architectures, datasets, or published benchmarks and do not compare their reported timings with ours.

POD-augmented conjugate gradients and subspace recycling already integrate data-derived spaces into linear-system solution [3]. Deflation theory links convergence to subspace quality and the accuracy of its numerical projections [4]. Neural Krylov Iteration further demonstrates that learning can be incorporated into Krylov acceleration [5]. These methods motivate stronger numerical controls; adding a learned basis would not by itself establish a distinct contribution here.

Recent work separates correctness, learning contribution, and complete numerical utility and demonstrates that initial-error summaries can misrank correction work [6]. This is directly relevant prior art, not a claim that our study originates that distinction. Reliability analyses of hybrid neural PDE solvers also emphasize physical residuals rather than merely vanishing learned updates [7]. We accordingly certify each returned solution using a freshly computed physical residual. Our source-linear architecture is an elementary structural regression model; its value, if any, must be empirical and limited to the declared family.

3 Problem formulation

On $[0, 1]$ we solve the dimensionless boundary-value problem

$$-\frac{d}{dx}\left(a(x)\frac{du}{dx}\right) = f(x), \qquad u(0) = u(1) = 0, \quad a(x) > 0. \tag{1}$$

The coefficient and source coordinates are $c, d \in \mathbb{R}^4$:

$$a(x) = \exp\left(\sum_{j=1}^{4} c_j \sin(j\pi x)\right), \qquad f(x) = \sum_{j=1}^{4} d_j \sin(j\pi x). \tag{2}$$

Independently sampled coordinates obey $c_j \sim U[-s, s]$ and $d_j \sim U[-1, 1]$. The training and IID value is $s = 0.45$; coefficient OOD uses $s = 0.9$. A distinct forcing shift adds independently sampled modes 5 through 8. The trained predictor retains its four-mode input interface; the correction algorithm always receives the complete forcing.

Let N be the total number of grid points, $h = 1/(N-1)$, and $n = N - 2$ the interior dimension. Arithmetic face coefficients $a_{i+1/2} = (a_i + a_{i+1})/2$ define the tridiagonal matrix

$$A_{ii} = h^{-2}(a_{i-1/2} + a_{i+1/2}), \qquad A_{i,i+1} = A_{i+1,i} = -h^{-2}a_{i+1/2}. \tag{3}$$

Training targets solve this discrete system. Thus error against a target measures discrete-solver agreement, not continuum discretization error. A separate manufactured-solution refinement test checks the expected second-order trend for a smooth constant-coefficient case.

Proposition 1 (Positive definiteness and classical spectral control)**.** *For nonzero v with zero boundary extension, $v^T A v = h^{-2}\sum_{i=0}^{N-2} a_{i+1/2}(v_{i+1} - v_i)^2 > 0$. If T is the tridiagonal matrix with diagonal 2 and adjacent entries -1, then*

$$a_{\min} v^T(h^{-2}T)v \le v^T A v \le a_{\max} v^T(h^{-2}T)v. \tag{4}$$

Consequently the symmetrically Laplacian-preconditioned matrix has condition number at most $a_{\max}/a_{\min}$.

Proof. Expand the tridiagonal quadratic form into squared adjacent differences. Bound every positive face coefficient between its minimum and maximum, then apply the generalized Rayleigh quotient. No learned component is involved. The bound is independent of grid resolution for fixed coefficient bounds; it can still be poor for large coefficient contrast. □

4 Predictors and numerical correction

4.1 Chronological training and inference

Training first generates coefficient/source samples and computes their discrete targets with Thomas elimination. For m samples, the design $X \in \mathbb{R}^{m\times p}$ and target matrix $Y \in \mathbb{R}^{m\times n}$ define

$$W = \arg\min_{W\in\mathbb{R}^{p\times n}} \|XW - Y\|_F^2 + \operatorname{tr}(W^T \Lambda W). \tag{5}$$

We solve the augmented least-squares system $[X; \sqrt{\Lambda}]W = [Y; 0]$ in double precision using one multi-output factorization. The ridge value is 10^{-4}. There are no gradient steps, epochs, online updates, or test-based normalization. The quadratic control weakens only its intercept penalty by 10^{-6}, matching S1's convention. All other feature coefficients receive the full penalty.

For $z = [c, d] \in \mathbb{R}^8$, the control feature map is

$$\phi_q(c, d) = [1, z, \{z_i z_j : 1 \le i \le j \le 8\}] \in \mathbb{R}^{45}. \tag{6}$$

The source-linear ablation and proposed feature maps are

$$\phi_1(c, d) = [1, c] \otimes d \in \mathbb{R}^{20}, \tag{7}$$

$$\phi_2(c, d) = [1, c, \{c_i c_j : 1 \le i \le j \le 4\}] \otimes d \in \mathbb{R}^{60}. \tag{8}$$

Inference computes $\hat{u} = \phi W$ on the interior and appends exact zero boundaries. The model has pn fitted scalar coefficients. A shuffled-label control uses the identical 60-feature architecture and training inputs with a fixed independently seeded permutation of target rows. It measures the importance of the learned input/target pairing with matched parameter count and fitting procedure. Because 45 and 60 differ, the quadratic/structured comparison does not isolate architecture from capacity.

Proposition 2 (Exact source superposition)**.** *For either source-linear feature family, fixed c, and scalars α, β, $\hat{u}(c, \alpha d + \beta e) = \alpha\hat{u}(c, d) + \beta\hat{u}(c, e)$ in exact arithmetic. In particular $\hat{u}(c, 0) = 0$.*

Proof. The coefficient-dependent factor of the tensor product is fixed. Distribute the tensor product over the linear source combination and multiply by the fixed fitted weight matrix. □

The proposition does not imply a uniform approximation bound, robustness to unseen coefficient contrast, or fewer Krylov iterations. Polynomial truncation can fail outside its training regime; unrepresented source frequencies must be recovered by numerical correction.

With p features and n outputs, fitting costs approximately $O(mp^2 + p^3 + mpn + p^2 n)$ and inference $O(pn)$, with $O(pn)$ model storage. S1 instead refactors normal equations separately for each output and ensemble member. S2's multi-output implementation is checked against the S1 quadratic predictor on a small identical problem; changing the feature family remains a scientific change, not merely an optimization.

4.2 Preconditioned conjugate gradients

All iterative methods use the same PCG implementation. The diagonal preconditioner is $M = \operatorname{diag}(A)$; the stronger reference preconditioner is $M = T$. A positive scalar factor in M does not change exact-arithmetic PCG iterates. Each online solve includes construction and factorization of its preconditioner. Tridiagonal factors are reused within that solve, but not silently amortized across distinct methods or cases.

Starting from x_0, compute $r_0 = b - Ax_0$, $z_0 = M^{-1}r_0$, $p_0 = z_0$. At step k use

$$\alpha_k = \frac{r_k^T z_k}{p_k^T A p_k}, \qquad x_{k+1} = x_k + \alpha_k p_k, \tag{9}$$

$$r_{k+1} = r_k - \alpha_k A p_k, \qquad z_{k+1} = M^{-1} r_{k+1}, \tag{10}$$

$$\beta_k = \frac{r_{k+1}^T z_{k+1}}{r_k^T z_k}, \qquad p_{k+1} = z_{k+1} + \beta_k p_k. \tag{11}$$

Nonfinite or nonpositive curvature ends the attempted iteration. The limit is $4n$ iterations. On apparent recursive-residual convergence, recompute $b - Ax$; if the true residual fails, restart PCG from it. Every output receives a final independent acceptance check

$$\|b - Ax\|_2 \le \max(10^{-12}, \tau\|b\|_2). \tag{12}$$

We retain the returned solution, recurrence history, restart count, operator applications, accuracy, timing, and acceptance flag. Failures remain in the denominator.

The physics-only initial field solves the constant-coefficient reference equation $x_0 = (h^{-2}\bar{a}T)^{-1}b$, where $\bar{a}$ is the mean face diffusivity. The direct baseline solves $Ax = b$ by Thomas elimination and applies the same final acceptance criterion. Its zero Krylov iteration count is not a basis for a meaningful relative-iteration comparison; only time and accuracy are compared with it.

5 Experimental methodology

5.1 Predecessor and successor boundaries

S1 uses 192 training cases, 48 calibration cases, a five-member bootstrapped quadratic ridge ensemble, a 65-point grid and diagonal PCG with a maximum of 63 iterations. Its development set has 64 IID and 64 coefficient-OOD cases. Its original frozen confirmatory configuration uses separate seeds and 128 cases in each test split. The original success rule requires at least 15% mean iteration reduction and a paired-bootstrap lower endpoint of at least 5%, with every proposed IID solve converged. Its calibration scores pool spatial points; this manuscript makes no distribution-free coverage claim from those intervals.

The development-only tolerance sweep reuses S1 development cases at $10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}$. It is diagnostic and does not change S1's frozen endpoint. S2 is a separate study with new seeds, different features, stronger baselines, fresh acceptance checks and complete timing.

5.2 Evaluation matrix

S2 crosses grids $N \in \{65, 129\}$, residual tolerances $\tau \in \{10^{-6}, 10^{-10}\}$, and IID, coefficient-OOD and forcing-OOD regimes. Nine methods are evaluated on every case: direct, zero/Jacobi, quadratic/Jacobi, zero/Laplacian, physics/Laplacian, quadratic/Laplacian, degree-one/Laplacian, structured/Laplacian and shuffled/Laplacian. This jointly tests the proposed predictor, its degree ablation, the value of learned pairing, the influence of preconditioning, resolution sensitivity, tolerance sensitivity and two specified distribution shifts.

Development uses three independently generated training/evaluation blocks, each with 192 training cases per grid and 12 evaluation cases per regime and grid. The two candidate feature degrees are specified before outcomes. Confirmation uses five blocks and 32 evaluation cases per regime and grid. A predictor is fitted separately at each grid, so the grid comparison tests scaling with refitting rather than zero-shot resolution transfer. No new benchmark or hyperparameter is introduced after confirmation.

The random seed base is 1,000,000 for development, 3,000,000 for confirmation and 9,000,000 for engineering smoke tests. Block r adds $100000r$ and grid index g adds $10000g$. Training uses offsets beginning at zero; regime index j uses offsets beginning at $1000(j + 1)$. Counts below 1000 and disjoint namespaces prevent sample overlap. The forcing-shift random stream adds 7,000,000 to its case seed; training-target shuffling uses the block/grid base plus 999. The complete generator inputs are retained with every case.

5.3 Primary endpoint and uncertainty

The prospective S2 primary cell is IID, $N = 129$, $\tau = 10^{-6}$. Within a block, each proposed/baseline pair has identical input and acceptance criterion. For case i the iteration reduction is

$$R_i = \frac{k_i^{\text{base}} - k_i^{\text{proposed}}}{\max(k_i^{\text{base}}, 1)}. \tag{13}$$

Average these within each training block, then average blocks equally. The primary claim passes only if the mean reduction is at least 15% and the 95% bootstrap lower endpoint is at least 5% against *both* zero/Laplacian and physics/Laplacian, and all compared solves converge. The conjunction prevents choosing the weaker baseline after observation. All other cells and comparisons are descriptive. A favorable subgroup does not replace the frozen primary claim.

The bootstrap resamples complete training-block means, with 10,000 confirmation repeats and fixed seeds. Five independent fitted models provide limited tail resolution. Intervals are therefore descriptive rather than a high-powered population guarantee; every block effect is retained. Cases sharing a fitted model are not counted as independent training repetitions. We do not report significance claims from the exploratory matrix. Timing uses the analogous paired relative reduction, which is distinct from a ratio of aggregate mean times.

5.4 Cost accounting and verification

Methods execute in a deterministic randomized order on each timing pass. Each case receives two development passes or three confirmation passes; its reported time is the median. The timed path includes matrix assembly, inference where relevant, preconditioner construction, correction, and independently recomputed residual acceptance. Training data generation, model fitting and model loading are excluded from online time and explicitly separated. Fitting time, parameter count, process CPU time, peak RSS, Python/NumPy versions and thread environment are retained. No training amortization is assumed.

The full source and configuration are frozen before protected execution. A confirmatory runner compares their digests with its freeze receipt and refuses changed inputs or an existing output directory. The verifier checks the complete case/method/tolerance matrix, hashes, regenerated inputs, model refits, replayed numerical solutions, true residuals, errors, timing medians and regenerated analysis. It does not infer external scientific replication from a same-code replay.

Numerical tests include independently assembled dense-system solutions, exact zero forcing, constant-coefficient one-step Laplacian PCG, source superposition, nonfinite rejection, a manufactured-solution refinement check, source-lock rejection and rehashed-corruption detection. The hosted runner executes the tests before admitting the full matrix. CPU and wall budgets are bounded; infrastructure failures are preserved separately from scientific negatives.

6 Results

6.1 S1: preserved failure and prospective confirmation

The original S1 development result is unchanged: all 64 IID and 64 coefficient-OOD cases require 63 PCG iterations under both learned and zero starts. The diagnostic tolerance sweep finds an IID benefit of 12.86% at 10^{-2}, but an OOD change of -4.26%; the advantage disappears or reverses at more stringent settings. This diagnostic is not substituted for the frozen endpoint.

The original frozen S1 confirmation yields 0.00% IID and 0.00% coefficient-OOD mean iteration reduction. Its recorded disposition is `NEGATIVE`. The source-linear successor is therefore not presented as a successful rerun of S1: it changes the predictor, preconditioning controls and evaluation protocol and uses disjoint data.

6.2 S2: primary outcome and cost

Table 1 reports the primary cell, with 160 paired cases across five independent training blocks. Table 2 gives paired effect estimates; a mean of paired ratios differs from a ratio of mean times. The predeclared joint iteration rule is satisfied, producing `SUPPORTED` for this narrow primary hypothesis. All 17280 successor condition-level rows are retained; 0 fail true-residual acceptance.

Method	Mean iterations	Mean time (ms)	Mean rel. error
Direct Thomas	0.000	0.448	0.00e+00
Zero + Jacobi	126.069	3.246	2.20e-10
Quadratic + Jacobi	126.963	3.360	6.05e-11
Zero + Laplacian	12.556	3.212	2.33e-09
Physics + Laplacian	11.556	3.393	3.94e-09
Quadratic + Laplacian	10.544	2.853	4.87e-09
Degree 1 + Laplacian	10.425	2.789	4.94e-09
Structured + Laplacian	9.131	2.518	6.75e-09
Shuffled + Laplacian	12.619	3.326	2.58e-09

Table 1: Frozen S2 primary cell: IID, 129 grid points, relative tolerance 10^{-6}. Online time includes assembly, inference, setup, correction and residual certification. Training and loading are excluded.

The direct method is 5.61 times faster than the proposed method by the ratio of mean online times. Therefore the evidence does not support replacing direct 1D solving with this learned hybrid for throughput. The positive iterative comparison remains meaningful only when the permitted baseline class is iterative and the offline learning cost is separately justified.

Baseline	Iteration reduction	95% block interval	Online saving
Zero + Laplacian	27.73%	[27.28, 28.22]%	21.72%
Physics + Laplacian	21.17%	[20.94, 21.50]%	26.07%
Quadratic + Laplacian	13.58%	[13.17, 14.00]%	11.69%
Shuffled + Laplacian	28.07%	[27.33, 28.99]%	24.47%
Direct Thomas	–	–	-463.63%

Table 2: Structured predictor plus Laplacian PCG versus each listed baseline. Negative online saving means slower. Direct solving has no meaningful relative-iteration metric. Five-block bootstrap intervals have limited resolution.

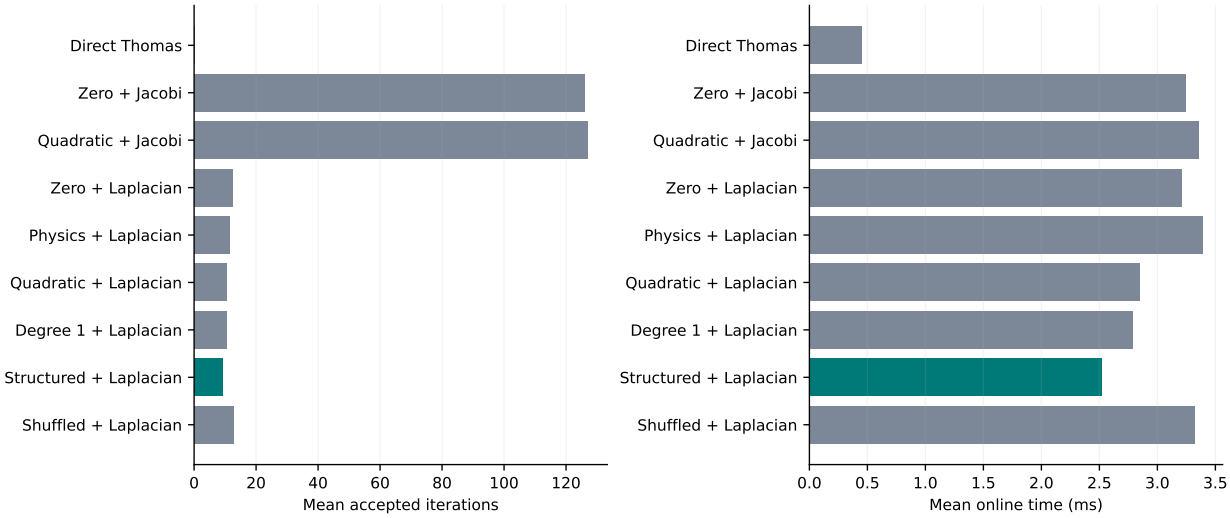


Figure 1: Accepted correction work and complete online time. The classical preconditioner changes the scale of correction work; direct tridiagonal solving remains the fastest measured path.

6.3 Ablation, sensitivity and distribution shifts

The degree-one and shuffled-label controls appear in Table 1. Both use the same PCG acceptance implementation. The shuffled control preserves the proposed feature count and fitting procedure while removing the input/target pairing. The degree ablation measures the additional coefficient-quadratic features; the unconstrained quadratic comparison also changes parameter count and cannot isolate structure alone.

Under the coefficient shift, the paired iteration reduction versus the physics baseline is 13.20%, with online saving 16.83%. Under the forcing shift, the paired iteration reduction versus the physics baseline is -2.29%, with online saving 8.61%. At the stricter IID tolerance 10^{-10}, the iteration reduction against the physics baseline is 12.80%. These are descriptive checks, not replacements for the primary endpoint. Figure 2 shows every grid/tolerance/regime comparison against both classical iterative starts; the appendix retains all method-level summaries.

6.4 Resources and evidence verification

The S2 producer records 175.59 CPU seconds, 175.75 wall seconds, and peak RSS 191.23 MiB before the subsequent verifier. These producer measurements exclude packaging, remote scheduling and later artifact replay. The runtime is Python 3.13.15 and NumPy 2.1.3 on Kaggle CPU with one BLAS/OpenMP thread. Offline fit and training-data generation measurements are retained per block, grid and model in `offline_costs.json`; they are not hidden inside an assumed amortization.

7 Discussion and limitations

The S1 observation distinguishes accurate correction from useful initialization. Convergence at the specified residual shows that the combined solver reaches an accurate discrete solution; it does not attribute that result to learning. At the strict endpoint, all observed methods exhaust the 63-dimensional iteration count, leaving no measured iteration

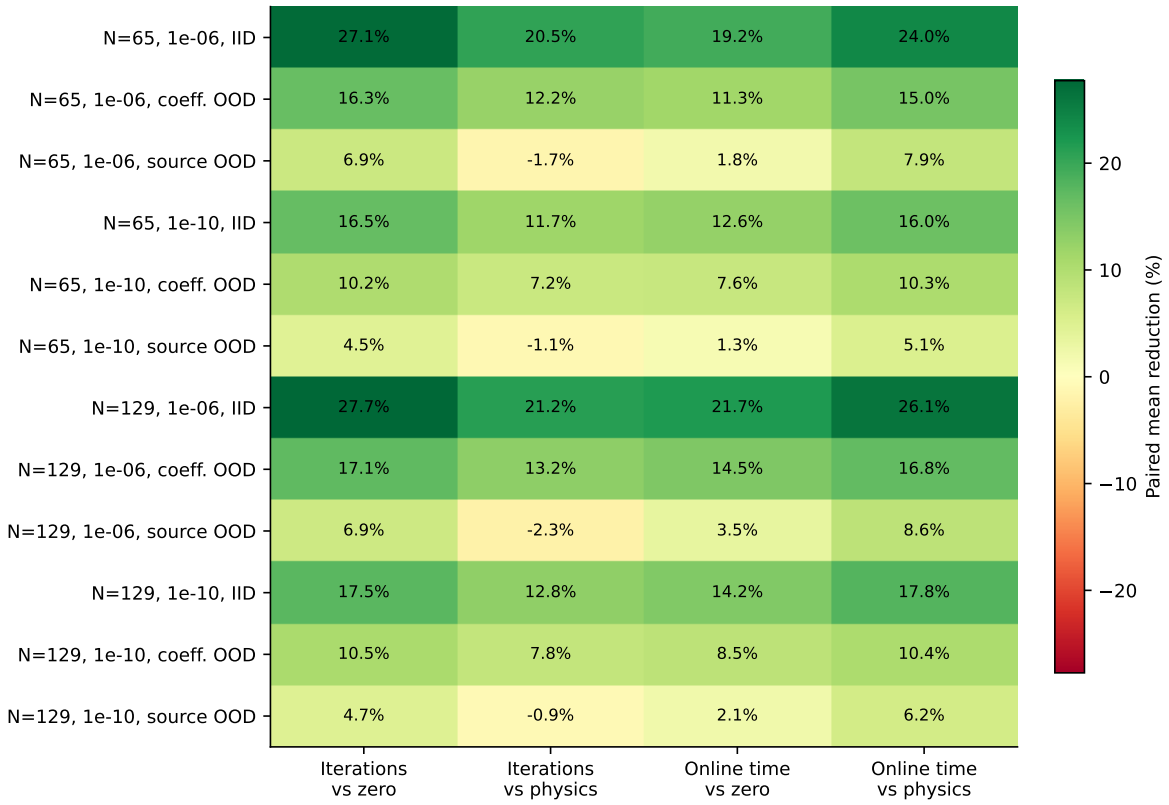


Figure 2: All S2 cells. Columns distinguish paired iteration and online-time reductions against zero and physics Laplacian starts. Negative values are retained.

advantage. This is consistent with small initial errors retaining a broad spectral support, but an observed count alone does not prove a unique spectral explanation.

Source linearity is a principled restriction that can be proved independently of experiment. Its computational value must still survive tests: a better predictor may not reduce accepted iterations, a reduction may not cover inference and setup costs, and an iterative path may remain slower than a direct method. The classical Laplacian condition-number bound also cautions against attributing gains caused by a preconditioner to the fitted predictor.

The study is deliberately narrow. Smooth four-coordinate coefficients, one-dimensional domains, modest grids and synthetic right-hand sides do not represent general PDE simulation. The forcing-shift test is limited to declared added modes. The regression model is not a deep neural operator. Five confirmation training blocks provide limited uncertainty resolution; randomized repeated timings remain host-specific. Offline training and model-loading costs are not amortized into a hypothetical workload. We make no state-of-the-art claim and no claim of independent replication by another researcher.

S1's pooled spatial calibration scores do not establish distribution-free conformal coverage for independent PDE instances. Those values remain in the archived evidence, but the coverage guarantee is withdrawn and the mechanism is omitted from S2. The source-linear predictor supplies a point estimate; certified numerical residual acceptance is a different property from probabilistic uncertainty calibration.

8 Reproducibility and research integrity

The repository preserves S1, a flaw register, source and protocol hashes, all admitted/failed attempts, raw synthetic inputs, trained coefficients, solver histories, timing passes and analysis. Confirmation is not used for subsequent tuning. A valid negative result closes the stated hypothesis; any new mechanism requires a successor version. Code replay and external scientific reproduction are reported separately. No human-subject data, personally identifying training data or proprietary external dataset is used.

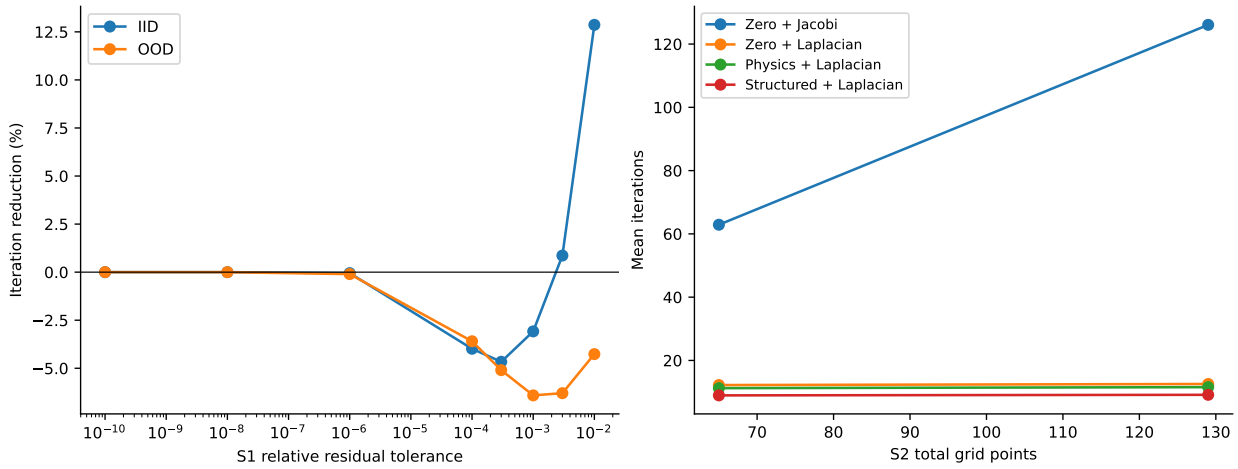


Figure 3: Left: the development-only S1 tolerance diagnostic. Right: IID S2 iteration scaling at 10^{-6} with separately fitted models per grid. The two panels concern different, explicitly versioned studies.

9 Conclusion

The frozen predecessor remains a negative result. The source-linear successor meets its predeclared iteration-reduction rule against both strong classical iterative starts on the specified IID cell. Complete online timing is reported separately, and the direct tridiagonal method remains faster. The resulting research claim is deliberately narrow: structural learning and preconditioning can interact usefully in this tested iterative setting, while a broader fastest-solver or multidimensional claim is unsupported. Preserved failures, fixed source identities and full raw artifacts make this conclusion inspectable and reproducible.

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A Implementation and artifact contracts

The source files `src/piml/s2.py` and `tests/test_s2.py` implement the successor and its numerical checks. Legacy S1 source, tests, configs and raw development evidence are preserved under `evidence/s1`; `SHA256.json`

binds those files byte for byte. The active legacy report formatter receives only a Python 3.11 syntax compatibility repair; the archived source is unchanged.

Each S2 run retains `config.json`, `source_manifest.json`, `environment.json`, `cases.jsonl`, `metrics.jsonl`, `models.json`, `offline_costs.json`, `analysis.json`, `receipt.json`, `progress.json`, a file manifest and verification receipt. An unsuccessful execution additionally retains a failure record and available partial measurements. Output directories are never overwritten.

B Reproduction commands

From the project directory, install the research extra and use one thread:

```
python -m pip install '.[research]'
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
export MKL_NUM_THREADS=1 PYTHONPATH=src
python -m unittest discover -s tests -v
python -m piml verify evidence/s1/development-S1-v1
python -m piml.s2 run configs/s2_development.json NEW_DEV_DIR
python -m piml.s2 verify NEW_DEV_DIR
```

Confirmatory execution additionally requires the frozen configuration and source receipt. Use a new output directory for an explicitly labeled replication; never present a second run as a new independent protected experiment. Exact delivery and artifact hashes are recorded in the release ledger rather than inferred from the presence of these commands.

C Adversarial checks and claim mapping

The flaw register covers endpoint saturation, direct-solver competitiveness, weak classical controls, prior art, structural misspecification, repeated factorization, recursive-residual drift, invalid uncertainty claims, timing order, training replication, multiplicity, semantic verification, provenance, packaging, resources, synthetic scope, shared reference code, unequal capacity, source drift and independent reproduction. A fix is accepted only when its stated evidence exists. The machine-readable claim ledger maps every retained numerical statement to its run, analysis and rendered table. Proofs establish only SPD, spectral equivalence and source superposition, not learned performance.

D Complete S2 matrix

N	τ	Regime	Method	Iterations	ms	Failures
65	1e-06	IID	Direct Thomas	0.00	0.245	0
65	1e-06	IID	Zero + Jacobi	62.89	1.630	0
65	1e-06	IID	Quadratic + Jacobi	62.97	1.716	0
65	1e-06	IID	Zero + Laplacian	12.22	1.778	0
65	1e-06	IID	Physics + Laplacian	11.24	1.888	0
65	1e-06	IID	Quadratic + Laplacian	10.21	1.596	0
65	1e-06	IID	Degree 1 + Laplacian	10.07	1.560	0
65	1e-06	IID	Structured + Laplacian	8.96	1.435	0
65	1e-06	IID	Shuffled + Laplacian	12.39	1.873	0
65	1e-06	coeff. OOD	Direct Thomas	0.00	0.248	0
65	1e-06	coeff. OOD	Zero + Jacobi	62.91	1.639	0
65	1e-06	coeff. OOD	Quadratic + Jacobi	63.00	1.727	0
65	1e-06	coeff. OOD	Zero + Laplacian	21.44	2.955	0
65	1e-06	coeff. OOD	Physics + Laplacian	20.51	3.078	0
65	1e-06	coeff. OOD	Quadratic + Laplacian	19.59	2.813	0
65	1e-06	coeff. OOD	Degree 1 + Laplacian	19.42	2.751	0
65	1e-06	coeff. OOD	Structured + Laplacian	18.19	2.642	0
65	1e-06	coeff. OOD	Shuffled + Laplacian	22.36	3.150	0
65	1e-06	source OOD	Direct Thomas	0.00	0.245	0
65	1e-06	source OOD	Zero + Jacobi	62.92	1.633	0
65	1e-06	source OOD	Quadratic + Jacobi	62.99	1.714	0

N	τ	Regime	Method	Iterations	ms	Failures
65	1e-06	source OOD	Zero + Laplacian	12.19	1.769	0
65	1e-06	source OOD	Physics + Laplacian	11.21	1.882	0
65	1e-06	source OOD	Quadratic + Laplacian	11.34	1.748	0
65	1e-06	source OOD	Degree 1 + Laplacian	11.36	1.723	0
65	1e-06	source OOD	Structured + Laplacian	11.32	1.732	0
65	1e-06	source OOD	Shuffled + Laplacian	12.30	1.868	0
65	1e-10	IID	Direct Thomas	0.00	0.247	0
65	1e-10	IID	Zero + Jacobi	63.00	1.634	0
65	1e-10	IID	Quadratic + Jacobi	63.00	1.719	0
65	1e-10	IID	Zero + Laplacian	18.74	2.610	0
65	1e-10	IID	Physics + Laplacian	17.74	2.712	0
65	1e-10	IID	Quadratic + Laplacian	16.91	2.444	0
65	1e-10	IID	Degree 1 + Laplacian	16.80	2.408	0
65	1e-10	IID	Structured + Laplacian	15.68	2.281	0
65	1e-10	IID	Shuffled + Laplacian	18.93	2.706	0
65	1e-10	coeff. OOD	Direct Thomas	0.00	0.249	0
65	1e-10	coeff. OOD	Zero + Jacobi	63.00	1.646	0
65	1e-10	coeff. OOD	Quadratic + Jacobi	63.00	1.736	0
65	1e-10	coeff. OOD	Zero + Laplacian	31.10	4.201	0
65	1e-10	coeff. OOD	Physics + Laplacian	30.19	4.319	0
65	1e-10	coeff. OOD	Quadratic + Laplacian	29.35	4.067	0
65	1e-10	coeff. OOD	Degree 1 + Laplacian	29.21	4.017	0
65	1e-10	coeff. OOD	Structured + Laplacian	28.12	3.893	0
65	1e-10	coeff. OOD	Shuffled + Laplacian	31.91	4.398	0
65	1e-10	source OOD	Direct Thomas	0.00	0.250	0
65	1e-10	source OOD	Zero + Jacobi	63.00	1.645	0
65	1e-10	source OOD	Quadratic + Jacobi	63.00	1.724	0
65	1e-10	source OOD	Zero + Laplacian	18.84	2.622	0
65	1e-10	source OOD	Physics + Laplacian	17.86	2.723	0
65	1e-10	source OOD	Quadratic + Laplacian	18.02	2.596	0
65	1e-10	source OOD	Degree 1 + Laplacian	18.02	2.562	0
65	1e-10	source OOD	Structured + Laplacian	17.99	2.581	0
65	1e-10	source OOD	Shuffled + Laplacian	18.96	2.717	0
129	1e-06	IID	Direct Thomas	0.00	0.448	0
129	1e-06	IID	Zero + Jacobi	126.07	3.246	0
129	1e-06	IID	Quadratic + Jacobi	126.96	3.360	0
129	1e-06	IID	Zero + Laplacian	12.56	3.212	0
129	1e-06	IID	Physics + Laplacian	11.56	3.393	0
129	1e-06	IID	Quadratic + Laplacian	10.54	2.853	0
129	1e-06	IID	Degree 1 + Laplacian	10.43	2.789	0
129	1e-06	IID	Structured + Laplacian	9.13	2.518	0
129	1e-06	IID	Shuffled + Laplacian	12.62	3.326	0
129	1e-06	coeff. OOD	Direct Thomas	0.00	0.453	0
129	1e-06	coeff. OOD	Zero + Jacobi	126.34	3.265	0
129	1e-06	coeff. OOD	Quadratic + Jacobi	126.96	3.372	0
129	1e-06	coeff. OOD	Zero + Laplacian	24.33	5.950	0
129	1e-06	coeff. OOD	Physics + Laplacian	23.33	6.103	0
129	1e-06	coeff. OOD	Quadratic + Laplacian	22.14	5.521	0
129	1e-06	coeff. OOD	Degree 1 + Laplacian	21.88	5.436	0
129	1e-06	coeff. OOD	Structured + Laplacian	20.41	5.129	0
129	1e-06	coeff. OOD	Shuffled + Laplacian	25.56	6.303	0
129	1e-06	source OOD	Direct Thomas	0.00	0.454	0
129	1e-06	source OOD	Zero + Jacobi	126.19	3.264	0
129	1e-06	source OOD	Quadratic + Jacobi	126.92	3.366	0
129	1e-06	source OOD	Zero + Laplacian	11.90	3.089	0
129	1e-06	source OOD	Physics + Laplacian	10.91	3.248	0
129	1e-06	source OOD	Quadratic + Laplacian	11.09	2.981	0
129	1e-06	source OOD	Degree 1 + Laplacian	11.09	2.951	0
129	1e-06	source OOD	Structured + Laplacian	11.04	2.961	0
129	1e-06	source OOD	Shuffled + Laplacian	12.03	3.206	0
129	1e-10	IID	Direct Thomas	0.00	0.454	0
129	1e-10	IID	Zero + Jacobi	127.00	3.309	0
129	1e-10	IID	Quadratic + Jacobi	127.00	3.411	0

N	τ	Regime	Method	Iterations	ms	Failures
129	1e-10	IID	Zero + Laplacian	19.48	4.826	0
129	1e-10	IID	Physics + Laplacian	18.49	5.036	0
129	1e-10	IID	Quadratic + Laplacian	17.46	4.450	0
129	1e-10	IID	Degree 1 + Laplacian	17.33	4.406	0
129	1e-10	IID	Structured + Laplacian	16.12	4.147	0
129	1e-10	IID	Shuffled + Laplacian	19.56	4.934	0
129	1e-10	coeff. OOD	Direct Thomas	0.00	0.464	0
129	1e-10	coeff. OOD	Zero + Jacobi	127.00	3.303	0
129	1e-10	coeff. OOD	Quadratic + Jacobi	127.00	3.398	0
129	1e-10	coeff. OOD	Zero + Laplacian	36.76	8.796	0
129	1e-10	coeff. OOD	Physics + Laplacian	35.76	8.980	0
129	1e-10	coeff. OOD	Quadratic + Laplacian	34.73	8.426	0
129	1e-10	coeff. OOD	Degree 1 + Laplacian	34.50	8.372	0
129	1e-10	coeff. OOD	Structured + Laplacian	33.09	8.074	0
129	1e-10	coeff. OOD	Shuffled + Laplacian	37.89	9.148	0
129	1e-10	source OOD	Direct Thomas	0.00	0.454	0
129	1e-10	source OOD	Zero + Jacobi	127.00	3.263	0
129	1e-10	source OOD	Quadratic + Jacobi	127.00	3.368	0
129	1e-10	source OOD	Zero + Laplacian	18.59	4.606	0
129	1e-10	source OOD	Physics + Laplacian	17.62	4.795	0
129	1e-10	source OOD	Quadratic + Laplacian	17.77	4.541	0
129	1e-10	source OOD	Degree 1 + Laplacian	17.77	4.492	0
129	1e-10	source OOD	Structured + Laplacian	17.70	4.497	0
129	1e-10	source OOD	Shuffled + Laplacian	18.75	4.740	0